Reverse genetic analysis of ANID_08533 in *Aspergillus nidulans*

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Introduction

- ANID_08533 is an uncharacterised gene in *Aspergillus nidulans*
- Bioinformatic annotation suggested a nucleoside related catalytic role
- Reverse genetics and diagnostic PCR were used to assess ANID_08533 deletion

Research Aim

- ➢ Investigate the possible function of ANID_08533
- ➢ Test whether ANID_08533 was replaced by Af-pyrG

Methods

- Transform deletion cassette into *A. nidulans*
- Purify putative transformants
- Extract genomic DNA
- Perform diagnostic PCR with two primer sets

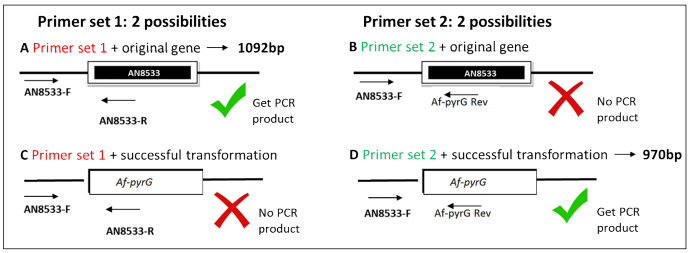


Figure 1. Expected diagnostic PCR outcomes for ANID_08533 deletion analysis. Primer set 1 detects the wild type locus (1092 bp), whereas primer set 2 detects the Af-pyrG junction in a correct deletion mutant (970 bp).

Results

Bioinformatic annotation

- ANID_08533 is an uncharacterised gene in *A. nidulans*
- Located on chromosome V, minus strand
- Encodes a 355 aa predicted protein
- Assigned to the nucleoside phosphorylase superfamily

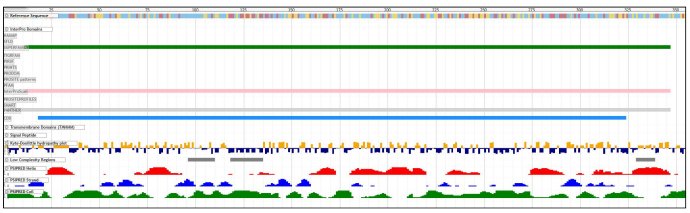


Figure 2. Protein feature and conserved domain annotation of ANID_08533. Domain predictions span most of the 355 aa sequence and support classification in the nucleoside phosphorylase superfamily. No obvious signal peptide or transmembrane domain was identified.

Expected diagnostic PCR outcomes

- **Primer set 1** detects wild type locus
  - ✓ Expected product: 1092 bp
- **Primer set 2** detects Af-pyrG junction
  - ✓ Expected product: 970 bp
- Correct deletion mutant
  - ➢ No product with primer set 1
  - ➢ 970 bp product with primer set 2

Putative transformants

- Original transformation plates from our group did not yield usable colonies
- Two laboratory provided putative transformants were purified on fresh agar
- Both remained viable after subculture and were used for diagnostic PCR

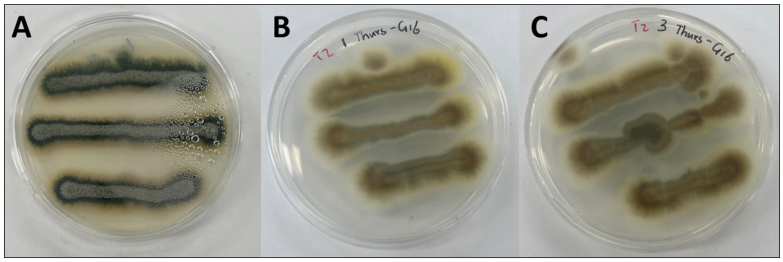


Figure 3. Purified colonies used for downstream analysis. A. original strain control; **B.** putative transformant 1; **C.** putative transformant 2. Both putative transformants remained viable after subculture.

Diagnostic PCR result

- No band in the water control
- Wild type sized bands were detected in lanes 5 and 7
- No clear mutant specific product was detected in lanes 6 and 8
- **Correct replacement was not confirmed**

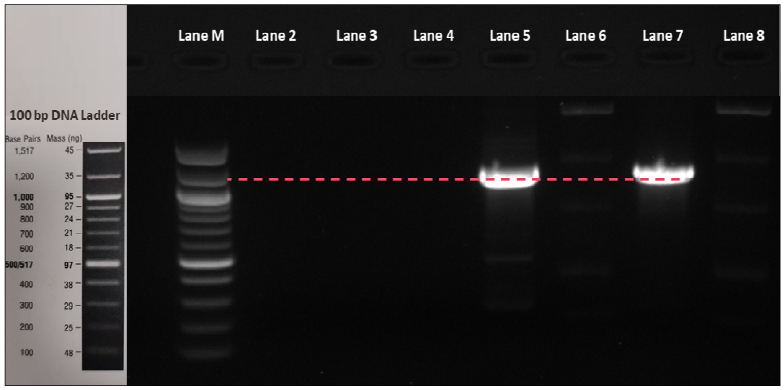


Figure 4. Diagnostic PCR gel. Lanes 5 and 7 showed WT-sized bands, while lanes 6 and 8 lacked mutant-specific Af-pyrG junction bands. Lane 3 WT control failed to amplify.

Conclusion

- ANID_08533 is predicted to encode a nucleoside related catalytic protein
- Correct Af-pyrG replacement was not confirmed by diagnostic PCR.
- Repeat PCR after single-colony purification is required

References

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